B8 Functions A level only

Name:

Class:

Date:

Time: **230 minutes**

Marks: **191 marks**

Comments:

Q1.

The function f is defined by $f(x) = e^{x-4}$, $x \in \mathbb{R}$

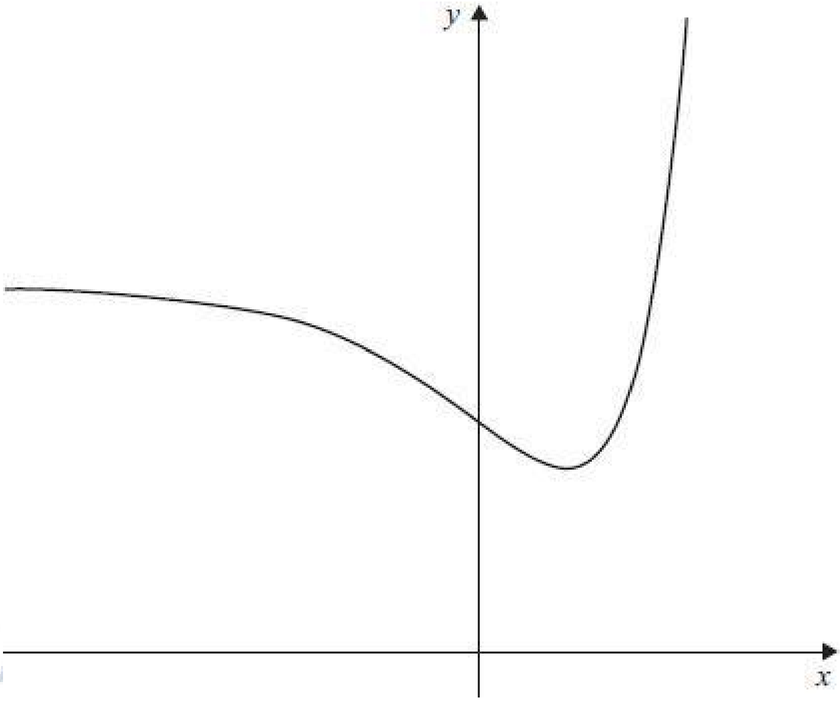
Find $f^{-1}(x)$ and state its domain.

(Total 3 marks)

Q2.

A function f has domain $\mathbb{R}$ and range $\{y \in \mathbb{R} : y \geq e\}$

The graph of $y = f(x)$ is shown.



The gradient of the curve at the point (x, y) is given by $\dfrac{dy}{dx} = (x - 1)e^x$

Find an expression for $f(x)$.

Fully justify your answer.

(Total 8 marks)

Q3.

The function f is defined by

$$f(x) = 4 + 3^{-x}, \quad x \in \mathbb{R}$$

(a) Using set notation, state the range of f

(2)

(b) The inverse of f is f^{-1}

(i) Using set notation, state the domain of f^{-1}

(1)

(ii) Find an expression for $f^{-1}(x)$

(3)

(c) The function g is defined by

$$g(x) = 5 - \sqrt{x}\ , (x \in \mathbb{R} : x > 0)$$

(i) Find an expression for $gf(x)$

(1)

(ii) Solve the equation $gf(x) = 2$, giving your answer in an exact form.

(3)

(Total 10 marks)

Q4.

The function f is defined by

$$f(x) = \frac{x^2 - 4}{3}, \text{ for real values of } x, \textbf{ where } x \leq 0$$

(a) State the range of f.

(2)

(b) The inverse of f is f^{-1}.

(i) Write down the domain of f^{-1}.

(1)

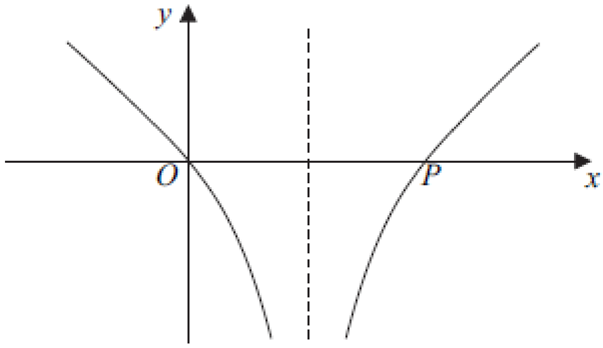
(ii) Find an expression for $f^{-1}(x)$.

(3)

(c) The function g is defined by

$$g(x) = \ln|3x - 1|, \quad \text{for real values of } x, \text{ where } x \neq \frac{1}{3}$$

The curve with equation $y = g(x)$ is sketched below.



(i) The curve $y = \text{g}(x)$ intersects the x-axis at the origin and at the point P.

Find the x-coordinate of P.

(2)

(ii) State whether the function g has an inverse. Give a reason for your answer.

(1)

(iii) Show that $\text{gf}(x) = \ln|x^2 - k|$, stating the value of the constant k.

(2)

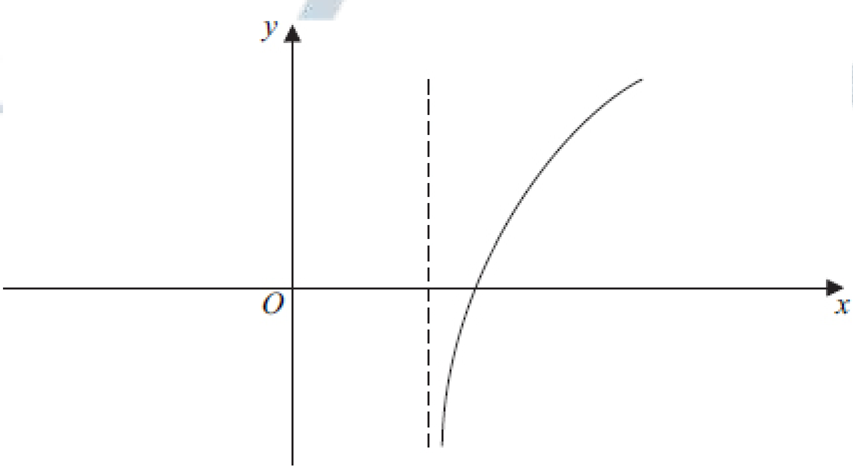
(iv) Solve the equation $\text{gf}(x) = 0$.

(4)

(Total 15 marks)

Q5.

The curve with equation $y = \text{f}(x)$, where $\text{f}(x) = \ln(2x - 3)$, $x > \frac{3}{2}$, is sketched below.



(a) The inverse of f is f^{-1}.

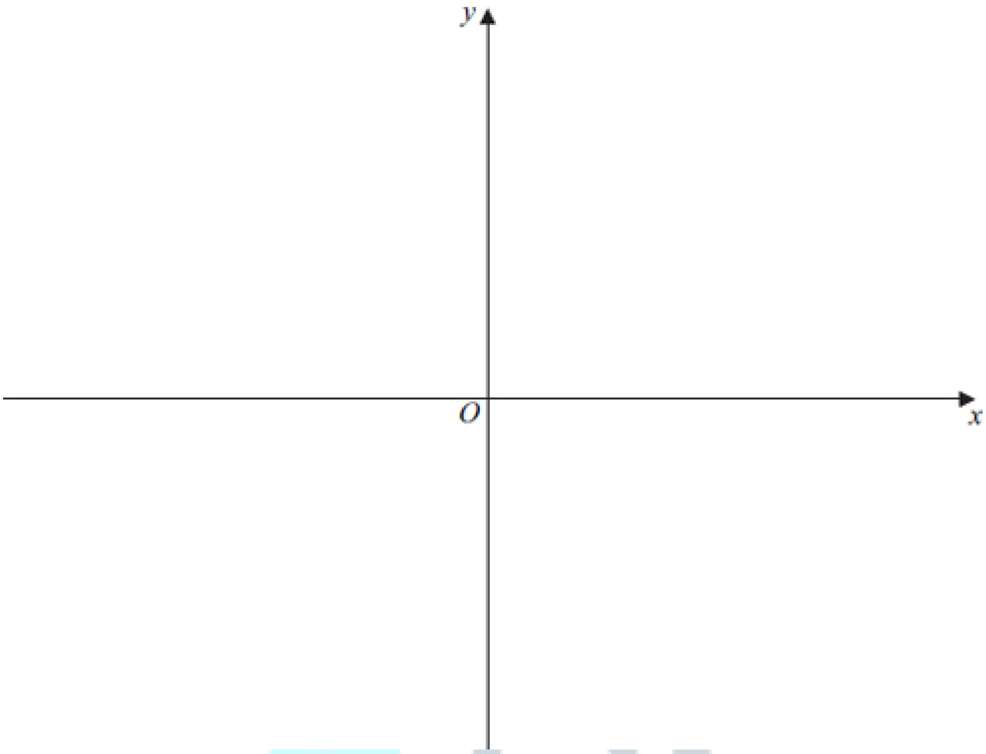
(i) Find $\text{f}^{-1}(x)$.

(3)

(ii) State the range of f^{-1}.

(1)

(iii) Sketch, on the axes given on the opposite page, the curve with equation $y = f^{-1}(x)$, indicating the value of the y-coordinate of the point where the curve intersects the y-axis.



(2)

(b) The function g is defined by

$$g(x) = e^{2x} - 4, \text{ for all real values of } x$$

(i) Find gf(x), giving your answer in the form $(ax - b)^2 - c$, where a, b and c are integers.

(3)

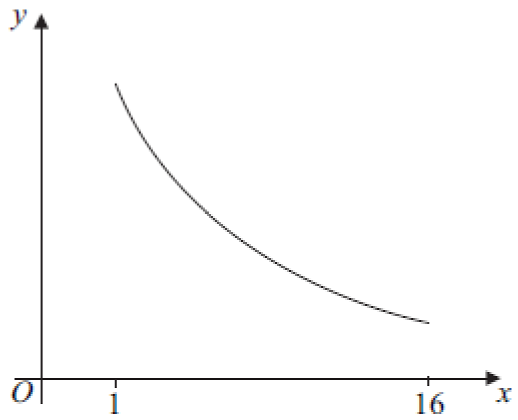
(ii) Write down an expression for fg(x), and hence find the exact solution of the equation fg(x) = ln 5.

(3)

(Total 12 marks)

Q6.

The curve with equation $y = \dfrac{63}{4x - 1}$ is sketched below for $1 \le x \le 16$.



The function f is defined by $f(x) = \dfrac{63}{4x-1}$ for $1 \le x \le 16$.

(a) Find the range of f.

(2)

(b) The inverse of f is f^{-1}.

(i) Find $f^{-1}(x)$.

(3)

(ii) Solve the equation $f^{-1}(x) = 1$.

(2)

(c) The function g is defined by $g(x) = x^2$ for $-4 \le x \le -1$.

(i) Write down an expression for $fg(x)$.

(1)

(ii) Solve the equation $fg(x) = 1$.

(3)

(Total 11 marks)

Q7.

The functions f and g are defined with their respective domains by

$f(x) = \sqrt{2x-5}$, for $x \ge 2.5$

$g(x) = \dfrac{10}{x}$, for real values of x, $x \ne 0$

(a) State the range of f.

(2)

(b) (i) Find $fg(x)$.

(1)

(ii) Solve the equation $fg(x) = 5$.

(2)

(c) The inverse of f is f^{-1}.

(i) Find $f^{-1}(x)$.

(3)

(ii) Solve the equation $f^{-1}(x) = 7$.

(2)
(Total 10 marks)

Q8.

The functions f and g are defined with their respective domains by

$f(x) = 3\cos\frac{1}{2}x$, for $0 \le x \le 2\pi$

$g(x) = |x|$, for all real values of x

(a) Find the range of f.

(2)

(b) The inverse of f is f^{-1}.

(i) Find $f^{-1}(x)$.

(3)

(ii) Solve the equation $f^{-1}(x) = 1$, giving your answer in an exact form.

(2)

(c) (i) Write down an expression for $gf(x)$.

(1)

(ii) Sketch the graph of $y = gf(x)$ for $0 \le x \le 2\pi$.

(3)

(d) Describe a sequence of two geometrical transformations that maps the graph of $y = \cos x$ onto the graph of $y = 3\cos\frac{1}{2}x$.

(3)
(Total 14 marks)

Q9.

The functions f and g are defined with their respective domains by

$f(x) = x^2$ for all real values of x

$$g(x) = \frac{1}{2x+1} \quad \text{for real values of } x,\ x \neq -0.5$$

(a) Explain why f does not have an inverse.

(1)

(b) The inverse of g is g^{-1}. Find $g^{-1}(x)$.

(3)

(c) State the range of g^{-1}.

(1)

(d) Solve the equation $fg(x) = g(x)$.

(3)
(Total 8 marks)

Q10.

The functions f and g are defined with their respective domains by

$$f(x) = e^{2x} - 3, \text{ for all real values of } x$$

$$g(x) = \frac{1}{3x+4}, \quad \text{for real values of } x,\ x \neq -\frac{4}{3}$$

(a) Find the range of f.

(2)

(b) The inverse of f is f^{-1}.

(i) Find $f^{-1}(x)$.

(3)

(ii) Solve the equation $f^{-1}(x) = 0$.

(2)

(c) (i) Find an expression for $gf(x)$.

(1)

(ii) Solve the equation $gf(x) = 1$, giving your answer in an exact form.

(3)
(Total 11 marks)

Q11.

(a) Find $\frac{dy}{dx}$ when:

(i) $y = \ln(5x - 2)$;

(2)

(ii) $y = \sin 2x$.

(2)

(b) The functions f and g are defined with their respective domains by

$f(x) = \ln(5x - 2)$, for real values of x such that $x \geq \frac{1}{2}$

$g(x) = \sin 2x$, for real values of x in the interval $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

(i) Find the range of f.

(2)

(ii) Find an expression for $gf(x)$.

(1)

(iii) Solve the equation $gf(x) = 0$.

(3)

(iv) The inverse of g is g^{-1}. Find $g^{-1}(x)$.

(2)

(Total 12 marks)

Q12.

The functions f and g are defined with their respective domains by

$f(x) = 2 - x^4$ for all real values of x

$g(x) = \frac{1}{x - 4}$ for real values of x, $x \neq 4$

(a) State the range of f.

(2)

(b) Explain why the function f does not have an inverse.

(1)

(c) (i) Write down an expression for $fg(x)$.

(1)

(ii) Solve the equation $fg(x) = -14$.

(3)

(Total 7 marks)

Q13.

The functions f and g are defined with their respective domains by

$f(x) = \sqrt{2x+5}$, for real values of x, $x \geq -2.5$

$g(x) = \dfrac{1}{4x+1}$, for real values of x, $x \neq -0.25$

(a) Find the range of f.

(2)

(b) The inverse of f is f^{-1}.

(i) Find $f^{-1}(x)$.

(3)

(ii) State the domain of f^{-1}.

(1)

(c) The composite function fg is denoted by h.

(i) Find an expression for $h(x)$.

(1)

(ii) Solve the equation $h(x) = 3$.

(3)

(Total 10 marks)

Q14.

The functions f and g are defined with their respective domains by

$f(x) = x^3$, for all real values of x

$g(x) = \dfrac{1}{x-3}$, for real values of x, $x \neq 3$

(a) State the range of f.

(1)

(b) (i) Find $fg(x)$.

(1)

(ii) Solve the equation $fg(x) = 64$.

(3)

(c) (i) The inverse of g is g^{-1}. Find $g^{-1}(x)$.

(3)

(ii) State the range of g^{-1}.

(1)

(Total 9 marks)

Q15.

The functions f and g are defined with their respective domains by

$f(x) = x^2$, for all real values of x

$g(x) = \dfrac{1}{2x-3}$, for real values of x, $x \neq \dfrac{3}{2}$

(a) State the range of f.

(1)

(b) (i) The inverse of g is g^{-1}. Find $g^{-1}(x)$.

(3)

(ii) State the range of g^{-1}.

(1)

(c) Solve the equation $fg(x) = 9$.

(3)

(Total 8 marks)

Q16.

The functions f and g are defined with their respective domains by

$f(x) = 3 - x^2$, for all real values of x

$g(x) = \dfrac{2}{x+1}$, for real values of x, $x \neq -1$

(a) Find the range of f.

(2)

(b) The inverse of g is g^{-1}.

(i) Find $g^{-1}(x)$.

(3)

(ii) State the range of g^{-1}.

(1)

(c) The composite function gf is denoted by h.

(i) Find $h(x)$, simplifying your answer.

(2)

(ii) State the greatest possible domain of h.

(1)

(Total 9 marks)

Q17.

The functions f and g are defined with their respective domains by

$$f(x) = \sqrt{x-2} \text{ for } x \geq 2$$
$$g(x) = \frac{1}{x} \text{ for real values of } x, x \neq 0$$

(a) State the range of f.

(2)

(b) (i) Find fg(x).

(1)

(ii) Solve the equation fg(x) = 1.

(3)

(c) The inverse of f is f^{-1}. Find $f^{-1}(x)$.

(3)
(Total 9 marks)

Q18.

The functions f and g are defined with their respective domains by

$f(x) = x^2$ for all real values of x

$g(x) = \dfrac{1}{x+2}$ for real values of x, $x \neq -2$

(a) State the range of f.

(1)

(b) (i) Find fg(x).

(1)

(ii) Solve the equation fg(x) = 4.

(4)

(c) (i) Explain why the function f does **not** have an inverse.

(1)

(ii) The inverse of g is g^{-1}. Find $g^{-1}(x)$.

(3)
(Total 10 marks)

Q19.

A function f is defined by $f(x) = 2e^{3x} - 1$ for all real values of x .

(a) Find the range of f.

(2)

(b) Show that $f^{-1}(x) = \frac{1}{3} \ln \left(\frac{x+1}{2}\right)$.

(3)

(c) Find the gradient of the curve $y = f^{-1}(x)$ when $x = 0$.

(4)
(Total 9 marks)

Q20.

The functions f and g are defined with their respective domains by

$f(x) = x - 2$ for all real values of x

$g(x) = \frac{6}{x+3}$ for all real values of x, $x \neq -3$

The composite function fg is denoted by h.

(a) Find $h(x)$.

(2)

(b) (i) Find $h^{-1}(x)$, where h^{-1} is the inverse of h.

(3)

(ii) Find the range of h^{-1}.

(1)
(Total 6 marks)